

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016–17

Zoology**Paper No. ZOB-505 : Environmental Biology & Systematics**

Time: Three hours

Max. Marks: 70

Note: Write your Roll No. at the top immediately on the receipt of this question paper. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

**Section A : Environmental Biology
(Marks : 45)**

Note: Answer four questions including Question 1 which is compulsory.

Question 1. Answer the following parts:

(a) State whether the following statements are true or false. Give brief justification: $[2 \times 5 = 10]$

- (i) Rain water is the major source of fresh water resource.
- (ii) Standing crop amounts living material in different tropic level.
- (iii) Energy flow is a multidirectional process.
- (iv) Predation is a kind of interaction.
- (v) Frequency, abundance and density are the qualitative characteristics of a community.

(b) Define the following: $[1 \times 5 = 5]$

- (i) Aquifers
- (ii) Keystone species
- (iii) Carrying capacity
- (iv) Land use planning
- (v) Ecological density.

Question 2. Describe the principle and methods of soil conservation. $[10]$

Question 3. Discuss different level of biodiversity and also write about the main threats to biodiversity. $[10]$

Question 4. Define air pollution. Describe its main sources, type of pollutants and its impact on the air pollution. $[10]$

Question 5. Write an essay on population characteristics. $[10]$

Section B : Systematics
(Marks : 25)

Note: Answer three questions including Question 1 which is compulsory.

Question 1. Answer the following parts:

(a) Define the following: [1 × 3 = 3]

(i) Tautonym

(ii) Gamma taxonomy

(iii) ICZN.

(b) State whether the following statements are TRUE or FALSE giving reasons in 2–3 lines:
[1 × 4 = 4]

(i) A paratype is a single specimen selected by the author of a species as its type.

(ii) Tribe comes between order and class of the taxonomic category.

(iii) Semicolon is always used in between author's name and the year.

(iv) Typification is the grouping of objects or information based on similarities.

Question 2. What do you understand by Zoological classification? Describe the various kinds of Zoological classification. [9]

Question 3. Discuss various molecular techniques which are employed to identify different species of animals. [9]

Question 4. Write notes on the following: [4.5 × 2 = 9]

(a) Morphological characters

(b) Biological species concept.

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